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TITLE: TELCOM CUSTOMER CHURN PREDICTION

SYSTEM DESIGN SPECIFICATION DOCUMENT

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1. Introduction

1.1 Purpose

The purpose of this Design Specification Document is to describe the technical design of the Telecom Customer Churn Prediction System. This document explains how the system will be implemented, including architecture, modules, data handling, algorithms, and interfaces.

1.2 Scope

My system is designed to:

- Analyze telecom customer data
- Identify patterns that lead to customer churn
- Build machine learning models to predict churn
- Assist telecom companies in reducing customer loss

2. Overall System Architecture

The system follows a modular architecture consisting of four major components:

1. Data Input Layer
2. Data Processing Layer
3. Machine Learning Layer
4. Visualization and Reporting Layer

3. Detailed System Design

3.1 Data Input Design

Data Source:

- CSV or Excel files containing telecom customer data
- Fields include:
    - CustomerID
    - Gender
    - Tenure
    - Monthly Charges
    - ContractType
    - Internet Service
    - Payment Method
    - Churn (Yes/No)

Input Method:

- Data loaded using Python (Pandas library)

3.2 Data Preprocessing Design

This stage prepares raw data for machine learning. Some of the activities carried out are like:

- Handling missing values
- Removing duplicates
- Encoding categorical variables
- Scaling numerical features
- Splitting dataset into training and testing sets

Some of the tools I used are:

- Python
- Pandas
- NumPy
- Scikit-learn

The outputs,

Clean dataset ready for modeling.

3.3 Machine Learning Model Design

Main objective of this machine learning model is to Predict whether a customer will churn or not.

Algorithms Used:

- Logistic Regression
- Decision Tree
- Random Forest

Training Process include:

1. Split dataset (80% training, 20% testing)
2. Train models
3. Tune hyperparameters
4. Compare model performance

Evaluation Metrics also include:

- Accuracy

- Precision
- Recall
- F1-Score
- Confusion Matrix

Output:

- Best performing model
- Churn predictions
- Performance report

3.4 Dashboard Design

The dashboard I will develop using Power BI.

Dashboard Features:

- Total Customers KPI
- Total Churned Customers
- Churn Rate (%)
- Churn by Contract Type
- Churn by Internet Service
- Churn by Tenure
- Filter by Gender, Contract, Payment Method

Purpose:

Allow non-technical stakeholders to understand churn patterns visually.

4. Data Design

4.1 Data Structure

My dataset contains:

- Numerical attributes (e.g., tenure, monthly charges)
- Categorical attributes (e.g., contract type, payment method)
- Target variable: Churn (Yes/No)

4.2 Data Storage

- My data is Locally stored in CSV format

5. User Interface Design

5.1 Developer Interface

5.2 End-User Interface

- I used Power BI for interactive dashboard
- Simple filtering and drill-down capabilities
- Clear visualization of churn insights

6. System Constraints

- Requires Python 3.x
- Requires libraries: Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn
- Requires Power BI Desktop
- Minimum 4GB RAM recommended
- Structured dataset required

7. Security and Data Protection Design

- Personally identifiable information must be anonymized
- Data must comply with Kenya Data Protection Act (2019)
- Restricted access to project files
- Backup copies maintained
- No unauthorized data sharing

8. Testing Strategy

Unit Testing:

- Test preprocessing functions
- Validate encoding and scaling

Model Testing:

- Evaluate prediction accuracy
- Verify confusion matrix results

Dashboard Testing:

- Confirm charts display correct values
- Test filters and interactions

9. Future Enhancements

- Real-time churn prediction
- Integration with telecom CRM systems

- Automated retraining of models

10. Conclusion

The Telecom Customer Churn Prediction System is designed using a modular architecture that ensures scalability, maintainability, and efficiency. The integration of machine learning models with Power BI dashboards enables data-driven decision-making to reduce customer churn.

APPENDICES

Glossary

This section defines technical terms, acronyms, and abbreviations used in the design specification document.

Churn
The event where a customer stops using the telecom service provider.

CustomerID
A unique identifier assigned to each customer in the dataset.

EDA (Exploratory Data Analysis)
The process of analyzing and summarizing datasets to understand patterns, trends, and relationships before building models.

Machine Learning (ML)
A method of data analysis that uses algorithms to automatically learn patterns from data and make predictions.

Logistic Regression
A classification algorithm used to predict binary outcomes (e.g., churn: Yes/No).

Random Forest
An ensemble machine learning algorithm that combines multiple decision trees to improve prediction accuracy.

Precision
The proportion of correctly predicted positive observations to the total predicted positives.

Recall
The proportion of correctly predicted positive observations to all actual positives.

F1-Score
The harmonic means of precision and recall.

Confusion Matrix
A table used to evaluate classification model performance by comparing predicted and actual values.

CSV (Comma-Separated Values)
A file format used to store tabular data.

Sample Dataset Structure

Below is a summary of key attributes in the dataset:

- customerID – Unique customer identifier
- gender – Male/Female
- tenure – Number of months customer has stayed
- PhoneService – Yes/No
- InternetService – DSL/Fiber optic/No
- Contract – Month-to-month/One year/Two year
- PaymentMethod – Electronic check/Mailed check/Bank transfer/Credit card
- MonthlyCharges – Monthly amount charged
- TotalCharges – Total amount charged
- Churn – Target variable (Yes/No)

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